PV102 Day 1 Student Packet

Applications and design decisions

Quiz | Reference | Design lab

Student Print Packet | Audience: Student | Monday, August 3, 2026

Today's rule

Start with load, site, and need. Record unknowns instead of guessing.

Learning targets

- Distinguish four common PV architectures.
- Map a first-pass energy path.
- Use client and site evidence to defend a design choice.
- State protection and verification boundaries before construction.

Morning quiz

1. What type of electrical output is produced directly by a PV module?

A. AC
B. DC
C. Mechanical
D. Thermal

2. When matching PV modules are connected in series, which quantity primarily adds?

A. Voltage
B. Current
C. Sunlight
D. Frequency

3. When matching PV strings are connected in parallel, which quantity primarily adds?

A. Voltage
B. Current
C. Resistance only
D. Frequency

4. For the small battery-based training systems used in PV101, which connection should be made first when the controller manufacturer's procedure requires battery-first startup?

A. PV to controller
B. Load to inverter
C. Battery to controller
D. Ground rod to roof

5. What is the main job of an overcurrent protective device such as a fuse?

A. Increase voltage
B. Limit damaging current
C. Create sunlight
D. Store energy

6. Before using continuity mode, the circuit must be:

A. Energized
B. De-energized and verified
C. Connected to the grid
D. Under full load

7. A 12 V load draws 2 A. What is its power in watts?

_____

8. Name two items that belong in a professional technical handoff.

_____

_____

9. A teammate is unsure whether a circuit is safe to touch. What is the correct next action?

_____

_____

10. In one sentence, explain why a correct component can still be wrong for a specific system.

_____

_____

Application architecture reference

Architecture	Utility role	Storage role	Typical decision signal
Grid-tied	Available and interconnected	Often none	Bill reduction is central; outage operation is not required
Hybrid	Available	Selected backup or operating goals	Grid benefits plus defined resilience needs
Off-grid	Unavailable	Carries loads outside solar hours	Remote loads must operate without utility support
PV-direct	Unavailable or irrelevant	Usually none	Load can operate only when solar power is available

Design order

1. Define the job: what must the system accomplish?
2. Define the load: what must run, at what power, and for how long?
3. Read the site: solar access, shade, weather, space, access, and maintenance.
4. Choose a first-pass architecture.
5. Map the energy path and name required protection and isolation points.
6. Record unknowns; verify them before sizing or building.

Safety boundary

Today's work is design-only. Use paper component cards or de-energized components. Do not connect modules, batteries, controllers, inverters, or utility circuits.

Afternoon design lab

Mission

Choose the PV architecture that best matches each client's real job. Show the energy path, identify the dominant constraint, state the safety boundary, and deliver a two-minute handoff.

Team structure

- Work in three teams of two. Start with the assigned scenario, then rotate if time permits.
- Roles: designer and verifier/recorder. Switch roles after each scenario.
- Use component cards or de-energized components only. No electrical connections are permitted.

Required deliverables

1. Architecture choice: grid-tied, hybrid, off-grid, or PV-direct.
2. A labeled energy-path diagram from source to load.
3. The critical load or operating need.
4. Three site or client constraints and the dominant constraint.
5. Protection, isolation, or labeling boundaries that must be designed before construction.
6. At least two unknowns that require verification.
7. A two-minute handoff: scope, decision, evidence, safety boundary, and unresolved items.

Acceptance criteria

Match the need. Show the path. Name the constraint. State the safety boundary. Record the unknowns.

Scenario 1 - Remote trail counter

- A 12 V electronic trail counter uses 8 W continuously.
- No utility service is available.
- The site has seasonal shade and can go five days between service visits.
- The counter must collect data day and night.

Design brief

1. Architecture choice and one-sentence reason

2. Critical load or operating need

3. Energy path - write components in order and draw arrows

4. Dominant constraint and supporting evidence

5. Two additional constraints

6. Required protection, isolation, and labeling boundaries

7. Unknowns to verify before sizing or building

8. Two-minute handoff notes

Scenario 2 - Small shop

- Utility service is available.
- The owner wants lower electric bills.
- During an outage, internet equipment and office lighting totaling about 400 W should operate for four hours.
- Roof area is available but one section receives afternoon shade.

Design brief

1. Architecture choice and one-sentence reason

2. Critical load or operating need

3. Energy path - write components in order and draw arrows

4. Dominant constraint and supporting evidence

5. Two additional constraints

6. Required protection, isolation, and labeling boundaries

7. Unknowns to verify before sizing or building

8. Two-minute handoff notes

Scenario 3 - Daytime water pump

- A 300 W DC pump fills an elevated tank.
- No utility service is available.
- The pump only needs to run when strong sunlight is available.
- The tank provides water after sunset; battery storage is not requested.

Design brief

1. Architecture choice and one-sentence reason

2. Critical load or operating need

3. Energy path - write components in order and draw arrows

4. Dominant constraint and supporting evidence

5. Two additional constraints

6. Required protection, isolation, and labeling boundaries

7. Unknowns to verify before sizing or building

8. Two-minute handoff notes

Peer verification and exit ticket

Peer verification checklist

☐ The architecture matches the client's stated operating need.

☐ The energy path includes every major required function.

☐ The dominant constraint is supported by scenario evidence.

☐ Protection and isolation are treated as design requirements.

☐ Unknowns are recorded instead of guessed.

☐ The handoff separates verified facts from open questions.

Exit ticket

1. Which operating need most strongly determines architecture?

2. When is storage unnecessary?

3. What is one question you must verify before Day 2 site assessment?